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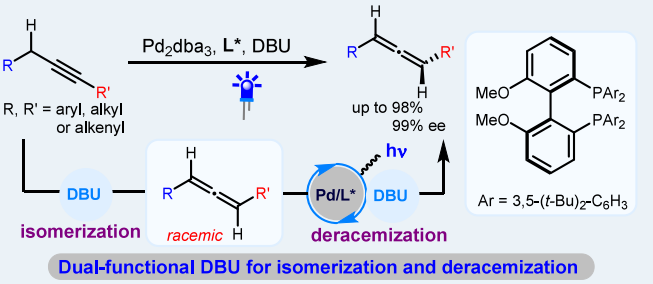
Photoexcited Palladium-Catalyzed Synthesis of Chiral Allenes from Alkynes via Isomerization and Deracemization Sequence

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ABSTRACT: A photoexcited palladium-catalyzed synthesis of chiral allenes from alkynes via a sequence of isomerization and deracemization is disclosed. This method offers an efficient and cost-effective approach to produce a wide range of chiral allenes with good enantioselectivities and yields (up to 98% yield and 99% ee). The use of readily available and stable alkynes as starting materials simplifies experimental procedures and broadens the scope of the chiral allene synthesis. DBU plays a crucial dual role in this reaction to effectively facilitate the isomerization of alkynes to allenes and enhance the palladium-catalyzed deracemization under visible light excitation, both of which are vital for the success of the process.



KEYWORDS: *alkynes, allenes, deracemization, isomerization, palladium*

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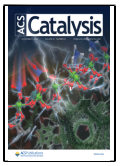
INTRODUCTION

Allenes are molecules with unique reactivity patterns and are widely utilized as building blocks for the synthesis of complex molecules, including natural products and bioactive compounds.[1] They can serve as enzyme inhibitors or antiviral agents,[2] and chiral allenes also function as ligands in asymmetric synthesis (Figure 1a).[3] Additionally, the axial-to-central chirality transfer in chiral allenes, which creates new stereogenic centers, has garnered significant interest among organic and medicinal chemists.[4] Despite the broad exploration of chiral allene applications, the asymmetric synthesis of these valuable molecules remains a challenging task.

Alkynes, as isomers of allenes, are frequently employed as starting materials for allene synthesis.[5] The isomerization of alkynes to allenes via 1,3-prototropic rearrangement under basic conditions remains one of the most direct and convenient methods.[5b] However, achieving enantioenriched chiral allenes through this strategy remains a significant challenge (Figure 1b). In 2000, Shioiri and co-workers reported an asymmetric isomerization of 1,3-diarylalkyne to axially chiral allene in the presence of a phase-transfer catalyst.[6] Although the enantioselectivity of the reaction was moderate, it represents an early example of constructing chiral allenes through alkyne isomerization. In 2009, Tan and co-workers demonstrated the Brønsted base-catalyzed isomerization reaction of alkynes to chiral disubstituted allenes with high enantioselectivity[7] where a bulky ester group was crucial for achieving this selectivity. In 2011, Takemoto and co-workers designed hydrogen-bond-donor catalysts that facilitated the enantioselective isomerization of alkynyl amides to their corresponding allenes.[8] More recently, in 2019, Kanai's group developed a Cu-catalyzed proton migration of skipped enynes to allenes with good regio- and enantioselectivity.[9] Despite these advances, significant challenges remain in the asymmetric isomerization of alkynes to chiral allenes: (1) the conversion rates are often suboptimal, and the similar polarity of alkynes and allenes complicates purification, and (2) substrates typically require preinstalled polar functional groups as binding sites, which limits the structural diversity. Therefore, there is a pressing need for a method that delivers high conversion rates and enantioselectivity, does not rely on specific functional groups, and is applicable to a broad range of substrates.

Very recently, our group successfully achieved the photoexcited palladium-catalyzed deracemization of nonprefunctionalized disubstituted allenes[10,11] by highlighting a significant base effect that enhances enantioinduction. This success, coupled with the fact that alkynes can isomerize to allenes under basic conditions,[5] has inspired us to explore the use of readily available and stable alkynes as starting materials for the one-pot synthesis of chiral allenes. In this process, we envision the base playing a dual role: facilitating the conversion of alkynes to racemic allenes and enhancing the stereoselectivity

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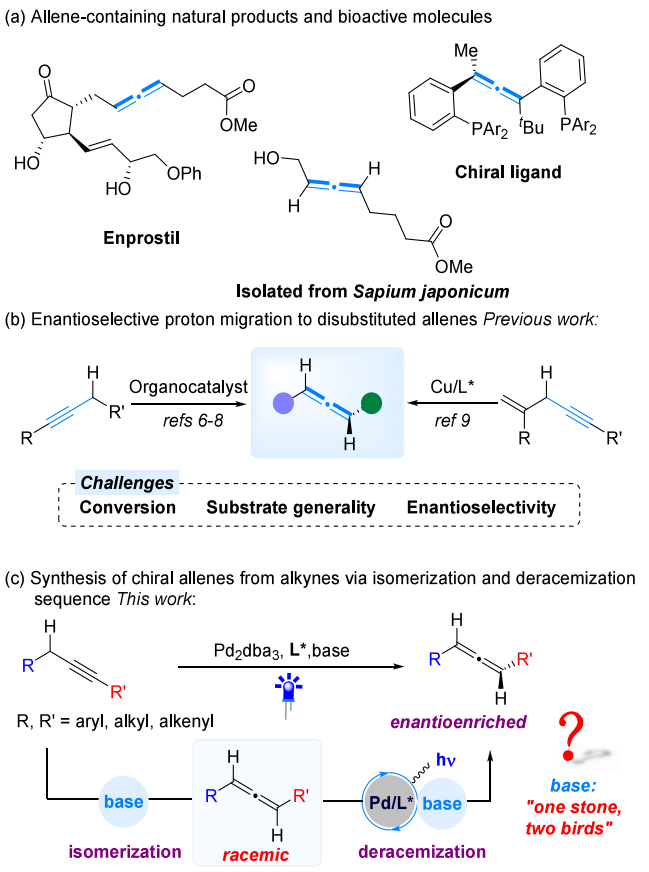


Figure 1. Allene-containing molecules and the synthesis of chiral allenes from alkynes.

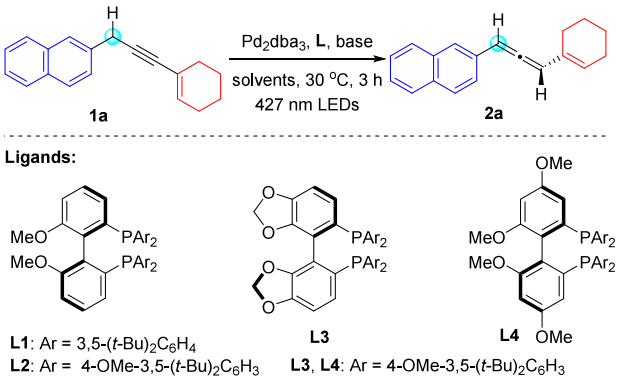
of the deracemization process (Figure 1c). Starting directly from alkynes, without the need to presynthesize racemic allenes, can significantly simplify experimental procedures and broaden the scope of chiral allene synthesis.

Ene-allenes are versatile building blocks in organic synthesis extensively used in electrocyclization and cycloaddition reactions.[1a−f] Chiral ene-allenes are particularly valuable as they enable the conversion of axial chirality to central chirality through cycloisomerization.[4b] However, the asymmetric synthesis of ene-allenes was rarely reported[9] and typically depended on chiral substrates as starting materials.[12] Recognizing the importance of chiral ene-allenes, we selected ene-alkyne **1a** as a model substrate to construct ene-allene **2a** using the conditions we recently developed for allene deracemization. Our goal of this study was to achieve the conversion of alkynes to allenes while enhancing the stereoselectivity of the deracemization process using a single base.

RESULTS AND DISCUSSION

Ene-alkyne **1a** was subjected to our optimized conditions for the allene deracemization.[10] A solution of **1a** (0.1 mmol), Pd_2dba_3 (2.5 mol %), **L1** (6 mol %) and K_2CO_3 in DMSO was irradiated with 40 W blue LEDs (λ_{max} = 427 nm) at 30 °C for 3 h (entry 1, Table 1). Unfortunately, the experimental results were not what we expected. The reaction proceeded slowly, resulting in low conversion (33%) and moderate enantioselectivity (77% ee). K_2CO_3 proved unsuitable as a base for this one-pot reaction. When KOH was then used as the base, alkyne **1a** was fully converted, but the resulting allene mostly

Table 1. Optimization of Reaction Conditions[a]



entry	L	base	solvent	conversion **1a**[b]	yield[b]	ee[c]
1	**L1**	K_2CO_3	DMSO	33	25	77
2	**L1**	KOH	DMSO	>99	9	32
3	**L1**	Et_3N	DMSO	0	0	
4	**L1**	DBU	DMSO	>99	>99	93
5	**L2**	DBU	DMSO	>99	>99	84
6	**L3**	DBU	DMSO	>99	>99	82
7	**L4**	DBU	DMSO	>99	>99	87
8	**L1**	DBU	DMA	>99	>99	85
9	**L1**	DBU	THF	90	87	34
10	**L1**	DBU	EA	>99	99	69
11[d]	**L1**	DBU	DMSO	>99	>99	84
12[e]	**L1**	DBU	DMSO	>99	>99	0
13		DBU	DMSO	>99	>99	0
14	**L1**		DMSO	0	0	

[a]Reaction conditions: a solution of **1a** (0.1 mmol), $Pd_2(dba)_3$ (2.5 mol %), ligand (6 mol %), and base (0.15 mmol) in the indicated solvent (2.0 mL) at 30 °C was irradiated by 40 W blue LEDs (427 nm) for 3 h. [b]Determined by GC. [c]Enantiomeric excess (ee) values were determined by HPLC on a chiral stationary phase. [d]Reacted with 456 nm LEDs instead of 427 nm LEDs. [e]Without light.

decomposed (entry 2). No product was detected when Et_3N was used (entry 3). Identifying a base that can both effectively promote the isomerization of alkynes to allenes and enhance the stereoselectivity of the deracemization process proved challenging. After extensive screening of bases (for more information, see Table S3 in the Supporting Information), we were pleased to find that DBU was the suitable choice. The chiral ene-allene **2a** was obtained in almost quantitative yield (>99% GC yield, 98% isolated yield) and excellent enantioselectivity (93% ee, entry 4). Other diphosphine ligands, such as MeO-BIPHEP (**L2**), SEGPHOS (**L3**), and GARPHOS (**L4**), did not yield better results (entries 5−7). The effects of various solvents were also explored. The use of dimethylacetamide (DMA), THF, and ethyl acetate (EA) resulted in unchanged reaction yields but significantly decreased enantiomeric excess (entries 8−10). Changing the wavelength of the light source to 456 nm did not affect the yield but reduced the enantiomeric excess (entry 11). In comparison to 427 nm LEDs, the emission spectrum of 456 nm LEDs exhibited less overlap with the absorption spectrum of the Pd/L/allene complex, resulting in a reduced deracemization efficiency. In the absence of light, palladium, or ligand, only racemic allenes were obtained (entries 12 and 13), suggesting that the reaction likely first produces racemic allenes, which then undergo deracemization to yield

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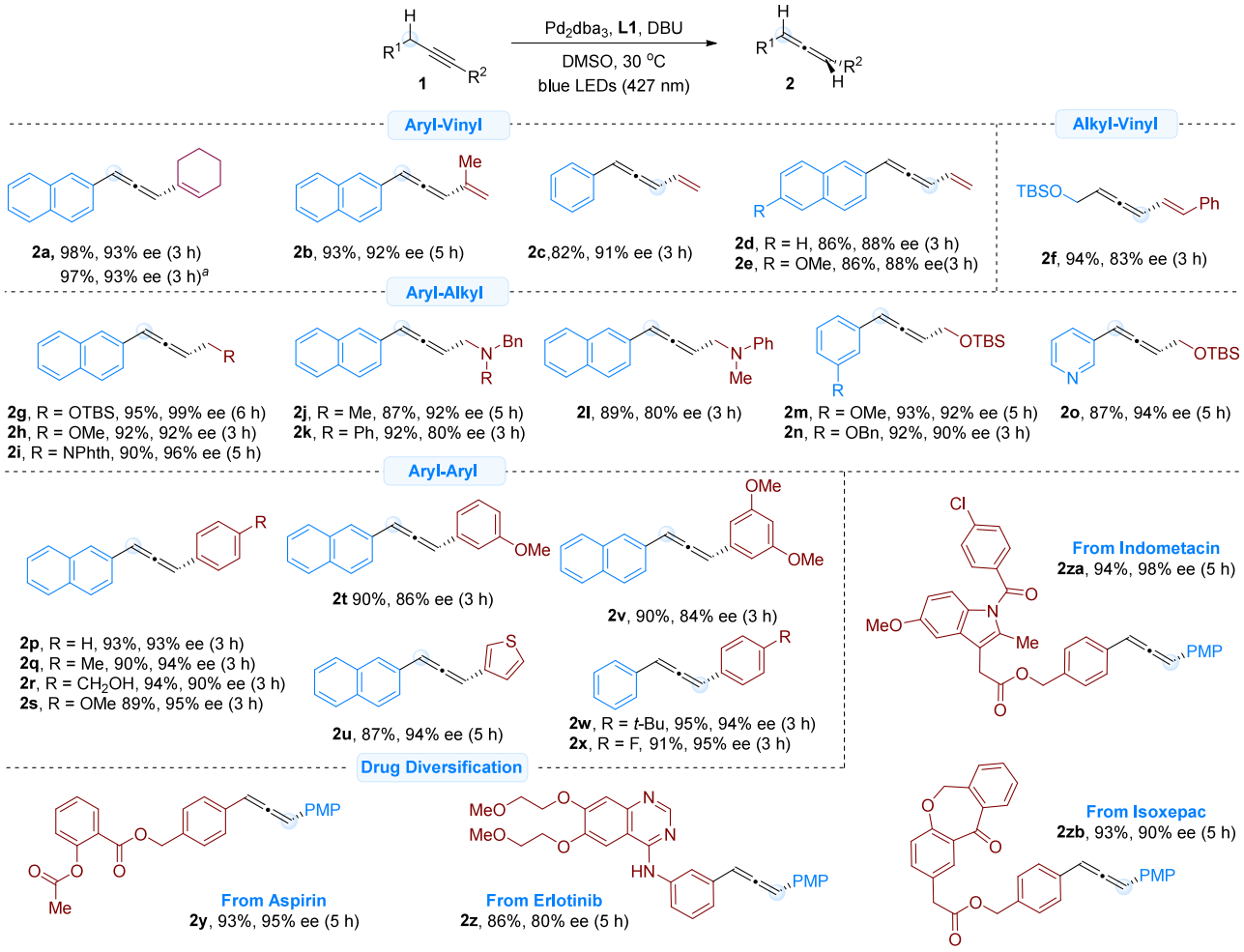


Figure 2. Substrate scope of the synthesis of allenes from alkynes. Reaction conditions: a solution of **1a** (0.1 mmol), $Pd_2(dba)_3$ (2.5 mol %), **L1** (6 mol %), and DBU (0.15 mmol) in DMSO (2.0 mL) at 30 °C was irradiated by 40 W blue LEDs (427 nm). [a]The reaction was scaled up to 4.5 mmol (**1a**).

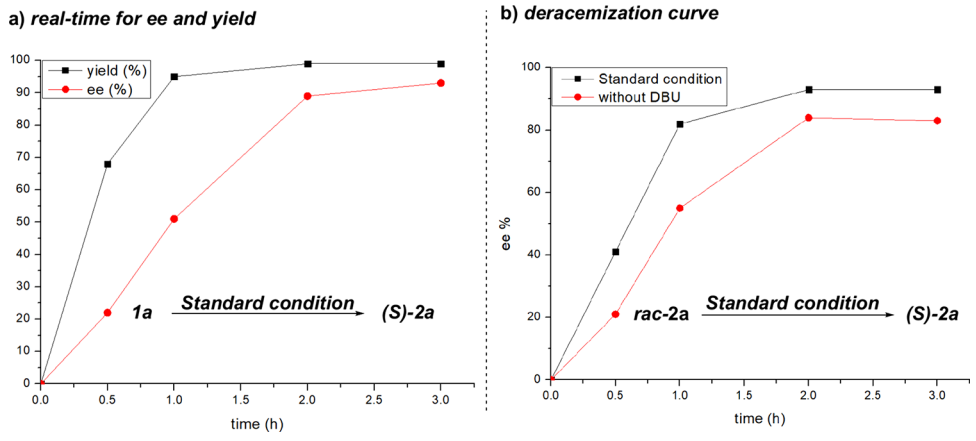


Figure 3. Kinetic experiments. (a) Real-time monitoring for ee and yield. (b) Time course profile for deracemization.

enantioenriched allenes. Without DBU, the alkyne was fully recovered, and no allene formation was detected (entry 14).

Having identified the optimal reaction conditions, we investigate the generality of our protocol (Figure 2). We were pleased to report that aryl-vinyl disubstituted allenes exhibit good yields (82−98%) and enantioselectivities (88−93% ee). Both internal (**2a**) and terminal (**2b**−**2e**) ene-alkynes yielded good results. Alkyl-vinyl disubstituted allenes (**2f**, 94%, 83% ee) were also compatible with our reaction system. Aryl-alkyl disubstituted allenes (**2g**−**2o**) provided good yields (87−95%) and good enantioselectivities (80−99% ee). Structures bearing alkoxy groups (**2g**, **2h**), tertiary amines (**2i**−**2l**), substituted phenyl rings (**2m**, **2n**), and aromatic heterocycles (**2o**) produced favorable results. Utilizing this strategy, aryl−aryl disubstituted allenes (**2p**−**2x**) also delivered excellent yields (87−95%) and stereoselectivities (84−95% ee).

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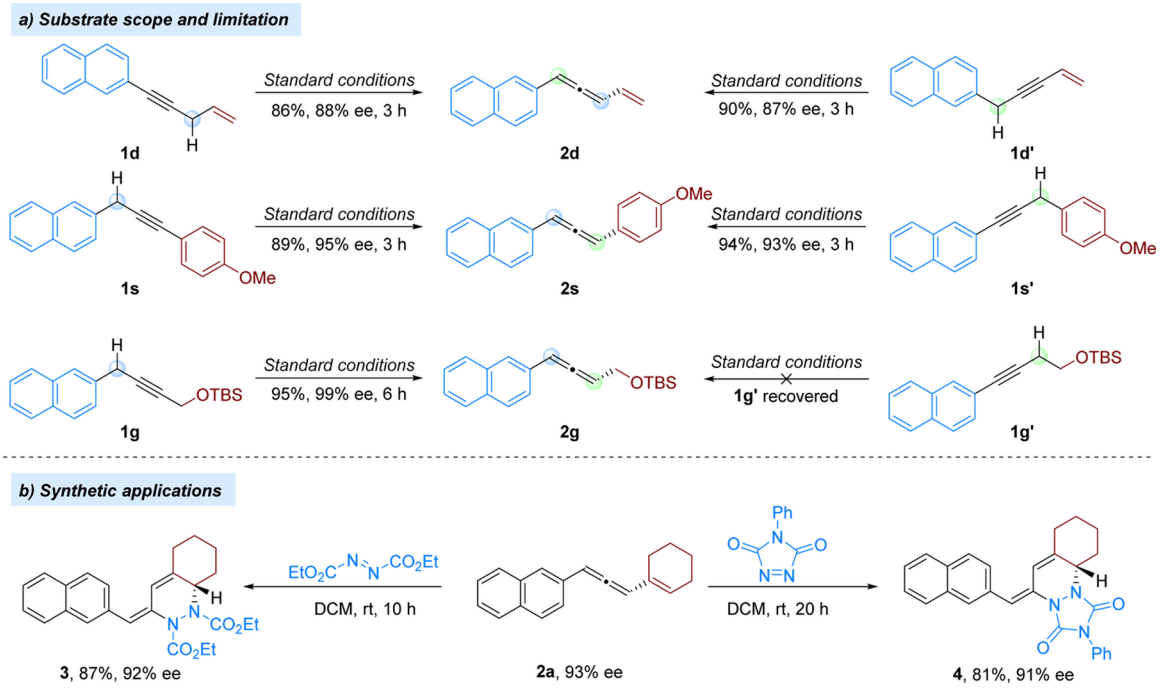


Figure 4. Practicality of the reaction.

Common functional groups, such as methyl (**2q**), unprotected hydroxyl (**2r**), bulky *tert*-butyl (**2w**), and halides (**2x**), were all compatible with this reaction system, demonstrating remarkable functional group tolerance. The heteroaromatic group, such as thiophene (**2u**), proved to be a competent reaction partner. Substituents at different positions (**2s**, **2t**) and number (**2v**) on the aryl rings of aryl−aryl disubstituted allenes were well tolerated. In exploring the broad utility of this procedure, we were delighted to observe that a series of complex substrates derived from aspirin (**2y**), erlotinib (**2z**), indometacin (**2za**), and isoxepac (**2zb**) could be transformed to the desired products with excellent stereoselectivities. The reaction could be scaled up to 4.5 mmol (**1a**) without affecting the yield (97%) and enantioselectivity (93% ee). Disappointingly, the synthesis of trisubstituted chiral allenes under standard conditions was unsuccessful (for more unsuccessful substrates, see Scheme S1 in the Supporting Information).

To validate the role of the base, we conducted kinetic experiments and monitored the reaction's progress (Figure 3). We observed that both the yield and enantiomeric excess (ee) of the allenes increased over time (Figure 3a). The presence of DBU significantly accelerates the conversion of alkyne **1a** to allenes **2a** with a notable 68% yield of **2a** achieved in just 30 min and full conversion to the allene occurring within 2 h. However, the rate of the allene deracemization lags behind the alkyne isomerization with only 22% ee detected at 30 min and its maximum being reached after 3 h. To further elucidate the function of DBU, we plotted the time-dependent deracemization curve of allene *rac*-**2a** (Figure 3b). Under standard conditions, the enantiomeric excess of the racemic allene peaked within 2 h. Additionally, excluding DBU from the reaction resulted in a decrease in both the rate and extent of deracemization. These findings suggest that DBU plays a dual role: promoting the conversion of alkynes to allenes and facilitating enantioenrichment of racemic chiral allenes, both of which are essential for achieving high yields and stereoselectivity of this one-pot reaction. (For a detailed possible reaction mechanism, see Scheme S3 in the Supporting Information).

To further demonstrate the practicality of this method and the versatility of substrates, we thoroughly investigated different alkyne isomers (Figure 4a). We found that aryl-alkenyl (**2d**) and aryl−aryl disubstituted allenes (**2s**) could be obtained with good yields and enantioselectivity, even when different alkyne precursors were used. Alkyne (**1g′**) with a weakly acidic hydrogen atom could not isomerize to allene (**2g**) under basic conditions, resulting in complete recovery of starting material **1g′**. The synthetic potentials of the alkenyl allenes (**2a**) were demonstrated via hetero-Diels−Alder reactions with diethyl azodicarboxylates and 4-phenyl-1,2,4-triazoline-3,5-dione (PTAD)[13] to afford bicyclic products **3** and **4** (Figure 4b). Throughout the conversion from axial to central chirality, the chirality was well maintained.

CONCLUSIONS

In summary, we have reported successful isomerization and deracemization from alkynes to chiral allenes with high yields and enantioselectivity. DBU exhibits a critical dual catalytic effect to efficiently facilitate the isomerization of alkynes into allenes and accelerate the palladium-catalyzed deracemization of allenes under visible light excitation, which is crucial for the success of this one-pot reaction. By employing a strategy of visible-light-induced palladium catalysis, a variety of disubstituted allenes, such as aryl-vinyl, alkyl-vinyl, aryl-alkyl, and aryl−aryl, have been successfully constructed. Utilizing stable and readily available alkynes as starting materials, we achieved an efficient asymmetric transformation, overcoming the challenges of low conversion rates and high substrate structure sensitivity that were prevalent in previous transformations. This method provides a new pathway for the synthesis of chiral allenes, characterized by its universality and broad adaptability.

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METHODS

General Procedure for Photoexcited Palladium-Catalyzed Synthesis of Chiral Allenes from Alkynes. In a nitrogen-filled glovebox, an 8 mL screw-cap test tube equipped with a magnetic stir bar was charged with $Pd_2(dba)_3$ (2.3 mg, 0.0025 mmol, 2.5 mol %), **L1** (6.2 mg, 0.006 mmol, 6 mol %), and anhydrous DMSO (1.0 mL), and the mixture was stirred for 10 min. Alkyne **1** (0.1 mmol), DBU (22.8 mg, 0.15 mmol), and anhydrous DMSO (1.0 mL) were added sequentially. The reaction tube was sealed with a Teflon screw cap and removed from the glovebox. The reaction mixture was stirred vigorously under blue LEDs (427 nm) at 30 °C (the temperature was maintained in a water bath) for the indicated reaction time. The reaction mixture was then transferred to a 100 mL separatory funnel, rinsed/diluted with 20 mL ether, and washed with 20 mL of deionized water (twice) and finally 20 mL of brine. The organic phase was concentrated under vacuum and purified by chromatography to afford chiral allene **2**.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.4c04956.

General experimental procedures and characterization, including ^{1}H NMR, ^{13}C NMR, and ^{19}F NMR spectral data (PDF)

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Notes

The authors declare no competing financial interest.

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